METRIC

A-A-59441/4

16 January 2004

COMMERCIAL ITEM DESCRIPTION SPECIFICATION SHEET

VAPOR CORROSION INHIBITORS, TYPE IV

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

The complete requirements for procuring Type IV vapor corrosion inhibitors (VCIs) described herein shall consist of this document and the latest issue in effect of A-A-59441.

1. SCOPE. This commercial item description covers the requirements for Type IV VCIs to provide corrosion protection for metals used in mechanical, electrical and electronic components and equipment that are subjected to corrosive environments. Type IV VCIs are suspended in liquids, which provide other or additional functions.

2. SALIENT CHARACTERISTICS

2.1 Physical characteristics. See Table I.

2.2 Performance. VCI material shall not interfere with or modify the electrical, optical, or mechanical surface properties of the treated components.

TABLE I. Physical characteristics.

| Material Form | Description | Application | Protection Life (mo) | Area Protected | Metals Protected |
|---|---|---|---|---|---|
| 1 | Oil-based VCI concentrate | Adds corrosion protection to lubricating or hydraulic oils. For use in indoor or open air applications. | 24 | Tanks, gear boxes, piping, etc. at 1 part VCI concentrate to 9 parts oil | MM |
| 2 | Oil-based VCI concentrate | Provides corrosion resistance for use with working engine systems. For use in indoor or open air applications. | 24 | Operating and layed-up engines at 1 part VCI concentrate to 9 parts oil | MM |
| 3 | Oil-based VCI concentrate | Corrosion inhibitor for gearboxes and hydraulic systems. | Up to 24 | Interior surfaces coated at 1l per $m^3$ (1 oz/$ft^3$) of enclosed space | MM |
| 4 | Solvent-based VCI rust preventative | General purpose corrosion protection for maintenance, lubrication, or temporary storage. | Up to 24 | Coated surface area (dip or spray) | MM |
| 5 | Oil-based VCI concentrate | Protection during lay-up periods of lubricating, hydraulic, and metal cutting systems. Metal parts storage. For use in sheltered conditions. | Up to 60 | Coated interior engine surfaces at following VCI ratios:<br>short-term 1 : 20<br>mid-term 1 : 10<br>long-term 1 : 5 | MM |



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TABLE I. Physical characteristics - Continued.

| Material Form | Description | Application | Protection Life (mo) | Area Protected | Metals Protected |
|---|---|---|---|---|---|
| 6 | Machining fluid, VCI, soluble oil | Protection of parts being machined or processed by automated machine tools. | Up to 24 | Coated parts (protection depends on concentration) | F & Zinc, Aluminum |
| 7 | Coating, VCI, temporary, wax-based | Protection of metal substrates exposed to harsh outdoor conditions. | Up to 3 | Coated parts at 50-7 0 µ (2-3 mils) | F & Aluminum, Copper |
| 8 | Coating, VCI, temporary, wax-based, ready-to-use | Same as 7 above. | Same as 7 above | Same as 7 above | F & Aluminum, Copper |
| 9 | Coating, VCI, temporary, oil-based, concentrate | Protection of metal substrates exposed to harsh outdoor conditions or mothballing. | 60+ | Same as 7 above | F & Copper, Aluminum, Magnesium, silver |
| 10 | Coating, VCI, temporary, oil-based, ready-to-use | Protection of metal substrates exposed to harsh outdoor conditions or mothballing. | 60+ | Same as 7 above | F & Copper, Aluminum, Magnesium, Silver |
| 11 | Coating, VCI, temporary, water-based, dry film | Protection of metal substrates for indoor storage (may be used for short-term outdoor exposure). | 36 | Coated parts at 75-125 µ (3-5 mils) | F |
| 12 | Coating, VCI, temporary, water-based, waxy film | Protection of metal substrates for indoor storage and outdoor exposure. | 60 (interior) 6-24 (exterior) | Coated parts at 5-50 µ (.2-2.0 mils) (Application-dependent) | F & Aluminum, Copper and Alloys, Tin |
| 13 | Coating, VCI primer, water-based, green color | Bonding other primers and coatings to metal substrates; replaces chromate treatments. | N/A | Coated parts at 12.5-25 µ (0.5-1.0 mils) | F |
| 14 | Coating, VCI primer, water-based, hydrophobic latex resin system, gray color | Bonding water-based topcoats to metal substrates. (Acceptable with selected solvent-based topcoats.) | N/A | Coated parts at 37.5-75 µ (1.5-3.0 mils) | F & Aluminum |
| 15 | Coating, VCI top-coat, water-based acrylic, non-skid, gray color | Non-skid, wear-resistant coating for floors. | Depends upon traffic | Coverage of 1.6-1.8 $m^2$/l (65-75 $ft^2$/gal) | F |



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TABLE I. Physical characteristics - Continued.

| Material Form | Description | Application | Protection Life (mo) | Area Protected | Metals Protected |
|---|---|---|---|---|---|
| 16 | Coating, VCI top-coat, water-based (specify color) | Top-coat or primer for harsh, outdoor applications -40 to 204 °C (-40 to 400 °F). | Varies with exposure | Coverage of 36 $m^2$/l (300 $ft^2$/gal) | F |
| 17 | Coating, VCI top-coat, water-based, gray or custom colors | Top coat for harsh atmospheres and UV resistance. Used with or without primers. | Varies with exposure | Coated parts at 325-375 µ (13-15 mils) | F |
| 18 | Cleaner and degreaser, water-based, alkaline, VCI, biodegradable | Cleaner which alters hydrocarbon deposits to permit removal by water and provides VCI protection. Conforms to ASTM D2240, ASTM F483-90, and ASTM F484-93. | 6 (Interior) | N/A | F |
| 19 | Cleaner and degreaser, water-based, alkaline, VCI, biodegradable, nitrite-free | Heavy duty cleaner for oil-field, commercial, and marine applications with VCI protection. | Varies with exposure | N/A | F |
| 20 | Organic rust remover, organic, VCI, non-toxic, non-flammable, biodegradable | Rust removal applications where protection of paints, plastics, wood, textiles, ceramics, or rubber is required. Outdoor and marine applications where disposal and pollution are problems. | Varies with exposure | N/A | Iron, Copper, Brass, Chrome, Aluminum |
| 21 | Rust remover, VCI, biodegradable, water-based, non-toxic, nitrite-free | Oxide scale removal and preparation of aluminum surfaces for further processing. | Varies with exposure | N/A | Aluminum |
| 22 | Concentrate, VCI, rust preventive, water-based, nitrite-free, phosphate-free | Water blasting, wet abrasive blasting, hydrotesting, fire sprinkler systems, or preservation. Suitable for saltwater applications. | Varies with exposure | N/A | F & Aluminum |
| 23 | Concentrate, VCI, nitrite-free | Corrosive solutions including closed loop cooling systems, fire extinguishing systems, tanks, cooling tower lay-up. | Continuous (or lay-up season) | 0.1 % (1,000 ppm) by volume (see manufacturer's recommendation for system lay-up) | MM |



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TABLE I. Physical characteristics - Continued.

| Material Form | Description | Application | Protection Life (mo) | Area Protected | Metals Protected |
|---|---|---|---|---|---|
| 24 | Additive, fuel system, VCI | Corrosion inhibitor, fuel stabilizer, and water emulsifier for gasoline, diesel, "gasohol", and alcohol fuels including methanol blends. | Continuous | 0.1-0.15 % by volume | MM |
| 25 | Additive, cooling system, VCI, nitrite-free, chromate-free | Corrosion inhibitor for ethylene glycol-based heat exchange fluids. | Continuous | 1.5-3.0 % by weight | F |
| 26 | Concentrate, metalworking, corrosion preventive, VCI, aqueous, nitrite-free, phosphate ester-free | Added to aqueous solutions for protection during metal working; may also be dipped or sprayed. Length of protection based on concentration (0.5-20%). | Varies with exposure | Added to metalworking fluids at 2 % by weight | F, Aluminum & Copper |
| 27 | Concentrate, corrosion preventive, VCI, aqueous | Diluted in water (5-50 %) and applied to finished metal parts. | Up to 24 (Indoor) | Varies | F, Aluminum & Copper |
| 28 | Corrosion inhibitor, VCI, distillate-based | Corrosion protection for internal metal parts during fabrication, storage or shipping. Used for fingerprint removal. | 24 (Indoor) | Coated parts at <2 µ | F, Aluminum & Copper |
| 29 | Grease, lithium complex, VCI | Lubrication and corrosion protection of metal-metal joints, bearings, sleeves and bushings. Corrosion inhibiting for salt water/brine, $H_2S$ and HCl environments. For both operation and Lay-up. | Varies with exposure | N/A | F |
| 30 | Additive, corrosion inhibiting, VCI | Additive for solvent-based, water-borne and epoxy coating systems. Also used with solvent-based waxes and solutions. | Varies with system | N/A | F & Aluminum |

Key: MM = Multi-metal; F = Ferrous; NF = Non-ferrous



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3. National stock numbers (NSNs). Table II provides NSNs that correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

TABLE II. National stock numbers.

| Material Form | Standard Package | NSN |
|---|---|---|
| 1 | 19 liter (5 gal) pails, 208 liter (55 gal) drums | TBD |
| 2 | 19 liter (5 gal) pails, 208 liter (55 gal) drums | TBD |
| 3 | 19 liter (5 gal) pails, 208 liter (55 gal) drums | TBD |
| 4 | 19 liter (5 gal) pails, 208 liter (55 gal) drums | TBD |
| 5 | 19 liter (5 gal) pails, 208 liter (55 gal) drums | TBD |
| 6 | 19 liter (5 gal) pails, 208 liter (55 gal) drums | TBD |
| 7 | 19 liter (5 gal) pails, 208 liter (55 gal) drums<br>312 gm (11 oz) aerosol cans | 8030-00-62-6950 |
| 8 | 19 liter (5 gal) pails, 208 liter (55 gal) drums | 8030-01-430-4898 |
| 9 | 19 liter (5 gal) pails, 208 liter (55 gal) drums<br>312 gm (11 oz) aerosol cans | 8030-00-244-1297 |
| 10 | 19 liter (5 gal) pails, 208 liter (55 gal) drums<br>312 gm (11 oz) aerosol cans | 8030-00-244-1296 |
| 11 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 12 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 13 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 14 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 15 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 16 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 17 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 18 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 19 | 2.3 kg (5 lb), 23 kg (50 lb), and 45 kg (100 lb) drums | TBD |
| 20 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 21 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 22 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 23 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 24 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 25 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 26 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 27 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 28 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |
| 29 | 400 gm (14.5 oz) cartridges, 19 liter (5 gal) pails, and 208 liter (55 gal) drums | TBD |
| 30 | 19 liter (5 gal) pails, 208 liter (55 gal) drums, and bulk | TBD |



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MILITARY INTERESTS

Custodians:
Navy – SH
Air Force – 99

Review Activities:
DLA – CC

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Navy – SH
(Project 2090-0118-004)



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